

12. During very cold weather, _____ of water helps in preserving the aquatic life.

- (a) contraction
- (b) expansion
- (c) evaporation
- (d) sublimation
- (e) None of the above / More than one of the above

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Ans. (b)

The expansion of water helps preserve aquatic life during very cold weather. When the temperature falls, the top layer of water in a pond contract becomes denser and sinks to the bottom. A circulation is set up until the entire water in the pond reaches its maximum density at 4°C. If the temperature falls further, the top layer expands and remains on the top till it freezes. Thus even though the upper layer are frozen the water near the bottom is at 4°C and the fishes etc. can survive in it easily.

13. The purest form of the amorphous carbon is

- (a) wood charcoal
- (b) sugar charcoal
- (c) bone charcoal
- (d) lampblack
- (e) None of the above / More than one of the above

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Ans. (b)

Sugar charcoal is the purest form of amorphous carbon. It is prepared by heating cane-sugar or glucose in the absence of air. It can also be prepared by dehydration of cane-sugar or glucose in the presence of sulphuric acid.

14. Which of the following is not a characteristic of a chemical change?

- (a) It is irreversible
- (b) New substance is formed
- (c) No net change in energy is involved
- (d) It involves absorption of energy
- (e) None of the above / More than one of the above

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Ans. (c)

A chemical change, often known as a chemical reaction, is the transformation of one or more compounds into new and distinct substances.

15. A mixture of sand and naphthalene can be separated by:

- (a) sublimation
- (b) distillation
- (c) chromatography
- (d) fractional distillation
- (e) None of the above / More than one of the above

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Ans. (a)

Naphthalene displays the phenomenon of sublimation, which is the transition of a substance directly from solid to gaseous medium upon heating. Hence, a mixture of sand and naphthalene can be separated by sublimation.

16. The term 'green chemistry' refers to–

- (a) a process of synthesizing chemical compounds by minimizing the use of hazardous chemicals
- (b) innovative scientific solutions to real-world environmental problems
- (c) the process that reduces the negative impact of chemical processes on human health and environment
- (d) More than one of the above

BPSC Agriculture Department-2024

Ans. (d)

Green chemistry is a chemical engineering and chemistry approach that aims to reduce or eliminate the use of hazardous substances in the creation of chemical products and processes. Green chemistry is also known as sustainable chemistry or circular chemistry. It is a process that reduces the negative impact of chemical processes on human health and the environment by designing chemical products and processes to reduce or eliminate the use of hazardous substances.

Inorganic Chemistry

1. The first periodic table was given by

- (a) Mendeleev
- (b) Dobereiner
- (c) Moseley
- (d) Newlands

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Ans. (d)

The first periodic table was given by Newlands but this table was based upon few elements and was therefore incomplete. The first comprehensive periodic table was given by Mendeleev. As per the question asked, the answer should be option (d.).

2. UNESCO inaugurated the celebration of 2019 as the International Year of the Periodic Table of Chemical Elements to celebrate its completion of how many years?

- (a) 100
- (b) 150
- (c) 75
- (d) 50
- (e) None of the above/More than one of the above

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Ans. (b)